Zaidan Muhammad Rafi 趙伊登

Electrical Power Engineering MSc Student | Wireless Power Transfer | Power Electronics
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SUMMARY

Master's student in Electrical Power Engineering at National Sun Yat-sen University (NSYSU), Taiwan, researching how to make **Wireless Power Transfer (WPT)** more efficient. Designed, built, and published a **140 kHz resonant WPT** prototype as an undergraduate, then worked as an **IoT NOC Engineer** handling live network and device incidents. **MOE Taiwan Scholarship** recipient, with a **journal paper, two patents, and WPT startup funding**.

Availability: internships during winter and summer school breaks; full-time roles after graduation in 2028.

EDUCATION

National Sun Yat-sen University (NSYSU) Sep 2026 - Aug 2028
MSc, International Master's Program in Electrical Power Engineering (IMEPE) *Kaohsiung, Taiwan*

- Research focus: improving the efficiency of Wireless Power Transfer systems.
- Courses: Power Electronic Converters, Power Conditioning Systems, Renewable Energy Applications.
- MOE Taiwan Scholarship, Taiwan Ministry of Education.

University of Brawijaya Aug 2021 - Jul 2025
BSc in Electrical Engineering, minor in Electronics | GPA 3.42/4.00 *Malang, Indonesia*

- Thesis: *Wireless Power Transfer for Low-Power IoT Devices* (140 kHz resonant WPT), published 2025.

TECHNICAL SKILLS

- **Power Electronics and WPT:** resonant WPT, DC-DC converters, PWM, half-bridge, coil design, LTspice
- **Hardware:** analog circuit design, PCB design and assembly, oscilloscope measurement
- **Embedded and Industrial:** Arduino, ESP32, SPI, PID control, Modbus TCP/IP, PLC, MQTT, LoRa
- **Software:** Python automation, network monitoring, Vue.js
- **Languages:** Indonesian (native), English (IELTS 6.5, TOEFL ITP 570), Mandarin (learning)

EXPERIENCE

NOC Engineer (IoT Specialist) May 2026 - Aug 2026
PT Geo Tekno Globalindo *Jakarta, Indonesia*

- Handled network and IoT device incidents, tracing root causes and getting systems back online fast.
- Worked remotely with on-board engineers to fix connectivity and network problems in the field.
- Built **Python tools** that automated parts of the team's NOC data analysis and cut manual work.

Electronics Project Intern Dec 2023 - Feb 2024
BRIN, Aerospace Research Organization (ORPA) *Bogor, Indonesia*

- Assembled a vibration monitoring and control system for a rocket, linking external ADC and DAC modules to an Arduino Mega 2560 over SPI.
- Added PID control to improve the system's response and the reliability of its data readings.

Research Assistant, Head of Smart Device Division Jul 2024 - May 2025
CDP Laboratory, University of Brawijaya *Malang, Indonesia*

- Developed smart device and WPT concepts, and helped secure **2+ research grants** for long-term WPT work.
- Prepared and taught a hands-on basic IoT workshop for **more than 150 high school students**.

Laboratory Coordinator Mar 2024 - May 2025
IoT and Smart Computing Laboratory, University of Brawijaya *Malang, Indonesia*

- Led a team of **six lab assistants** and managed the lab's staffing, equipment, activities, and external relations.
- Ran the 2024 assistant recruitment and led the revision of the practicum modules, including LoRa.
- **Promoted** from Laboratory Assistant (Mar 2023 - Mar 2024).

RESEARCH AND PROJECTS

Efficiency-Focused WPT Research Sep 2026 - Present
Master's research, NSYSU *Kaohsiung, Taiwan*

- Studying how to raise WPT efficiency, with **laptop-class (65 W) charging** as the target application.

Magnetic Resonant WPT for Low-Power IoT Devices Mar 2025 - Jun 2025
Undergraduate thesis, University of Brawijaya *Malang, Indonesia*

- Built a 140 kHz parallel-parallel resonant WPT system with a PWM-driven half-bridge and hand-wound coils.
- Tuned the resonance and measured performance across loads on real hardware; **published in 2025**.

Temperature Display via Ethernet and Modbus Jun 2024 - Jul 2024
Project for PT INKA (Indonesian railway manufacturer) *Indonesia*

- Built a Modbus TCP/IP prototype that shows PLC temperature and humidity on an LED matrix in train cars.

WaterGuard: Smart Fish Pond Monitoring Jul 2024 - Mar 2025
Team lead, community service project *Jember, Indonesia*

- Led a **six-person team** building a solar-powered IoT buoy that tracks turbidity, temperature, and TDS.

PUBLICATIONS AND PATENTS

- **Journal (2025):** Design and Development of a 140 kHz Magnetic Resonant Coupling WPT System for IoT Devices. Read the paper
- **Patent, Hydrozen (2024):** IoT monitoring and automatic temperature control for aquariums.
- **Patent, PIKO (2023):** waste processing with pyrolysis, RDF-based incineration, and electrical monitoring.

SELECTED AWARDS AND FUNDING

Brawijaya Start-up Action Funding Dec 2024
University of Brawijaya. Startup funding for the eight-member WPT team I led, focused on commercial WPT.

3rd Place, SIGMA Business Plan Competition Nov 2024
Telkom University. AI-based smart farming business plan for B2B and B2C customers.

2nd Place, Transport Safety Innovation Competition Jul 2024
Universitas Negeri Surabaya (UNESA). National innovation competition, occupational health and safety category.

PKM Student Research Funding Jun 2023
Indonesian Ministry of Education and Research. Research proposal selected for national funding.

Other awards: PMW Entrepreneurship funding (2024); 1st Place, PKM Dekan Cup, Faculty of Engineering (2022).

LEADERSHIP AND ACTIVITIES

Manager of Media and Publication Jul 2022 - Mar 2025
Internet of Things Universitas Brawijaya (student organization) *Malang, Indonesia*

- Relaunched Broditech, the organization's business unit, and led building the lab and organization websites.
- Earlier, as R&D staff, co-developed an IoT smart electric socket controlled from a mobile app.